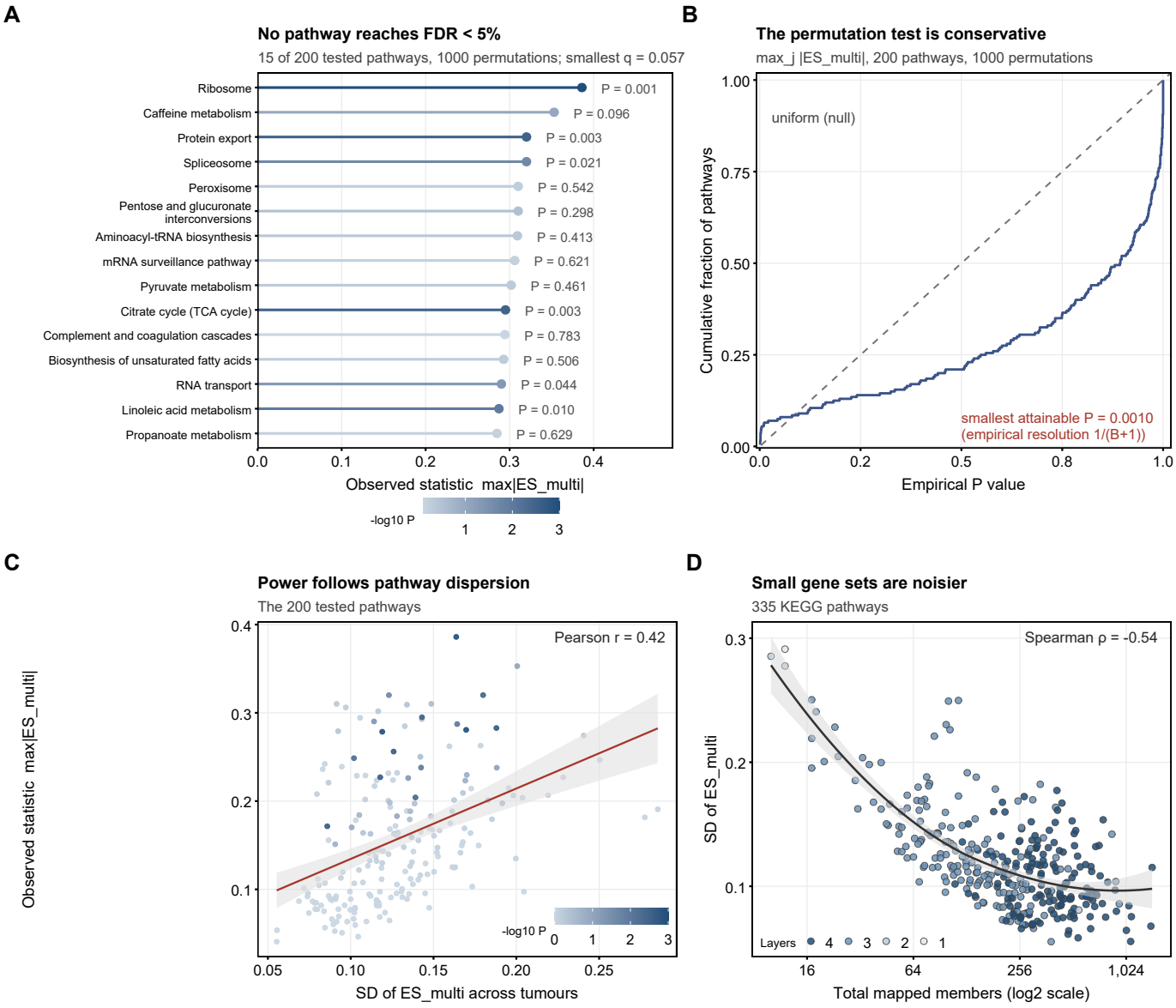

A
No pathway reaches FDR < 5%
15 of 200 tested pathways, 1000 permutations; smallest q = 0.057
Ribosome
Caffeine metabolism
Protein export
Spliceosome
Peroxisome
Pentose and glucuronate interconversions
Aminoacyl-tRNA biosynthesis
mRNA surveillance pathway
Pyruvate metabolism
Citrate cycle (TCA cycle)
Complement and coagulation cascades
Biosynthesis of unsaturated fatty acids
RNA transport
Linoleic acid metabolism
Propanoate metabolism
P = 0.001
P = 0.096
P = 0.003
P = 0.021
P = 0.542
P = 0.298
P = 0.413
P = 0.621
P = 0.461
P = 0.003
P = 0.783
P = 0.506
P = 0.044
P = 0.010
P = 0.629
0.0
0.1
0.2
0.3
0.4
Observed statistic max|ES_multi|
-log10 P
1
2
3
B
The permutation test is conservative
max_j |ES_multi|, 200 pathways, 1000 permutations
uniform (null)
smallest attainable P = 0.0010
(empirical resolution 1/(B+1))
Cumulative fraction of pathways
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.5
0.8
1.0
Empirical P value
C
Power follows pathway dispersion
The 200 tested pathways
Pearson r = 0.42
Observed statistic max|ES_multi|
0.4
0.3
0.2
0.1
-log10 P
0
1
2
3
0.05
0.10
0.15
0.20
0.25
SD of ES_multi across tumours
D
Small gene sets are noisier
335 KEGG pathways
Spearman ρ = -0.54
SD of ES_multi
0.3
0.2
0.1
Layers
4
3
2
1
16
64
256
1,024
Total mapped members (log2 scale)